9/11/2026 - MONTE CARLO, TEMPORAL DIFFERENCE

RECAP: DYNAMIC PROGRAMMING (P KNOWN), POLICY ITERATION

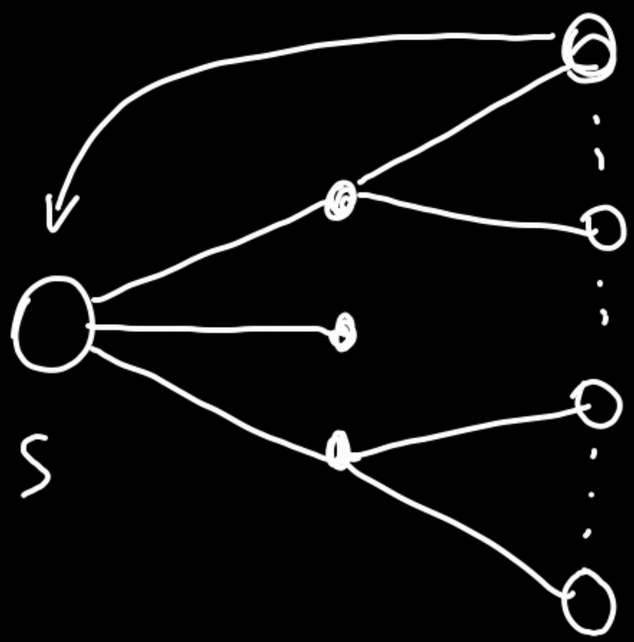
π INITIAL POLICY

POLICY EVALUATION ($\pi \to v_\pi$) - PREDICTION PROBLEM

$$v_\pi^{k+1}(s) = \sum_a \pi(a|s) \sum_{s',r} p(s',r|s,a)\left[r + \gamma v_\pi^k(s')\right], \forall s$$

EXPECTED UPDATE

BOOTSTRAPPING METHOD



$$v_\pi^k \xrightarrow[k\to\infty]{} v_\pi$$

$$q_\pi^k \xrightarrow[k\to\infty]{} q_\pi$$

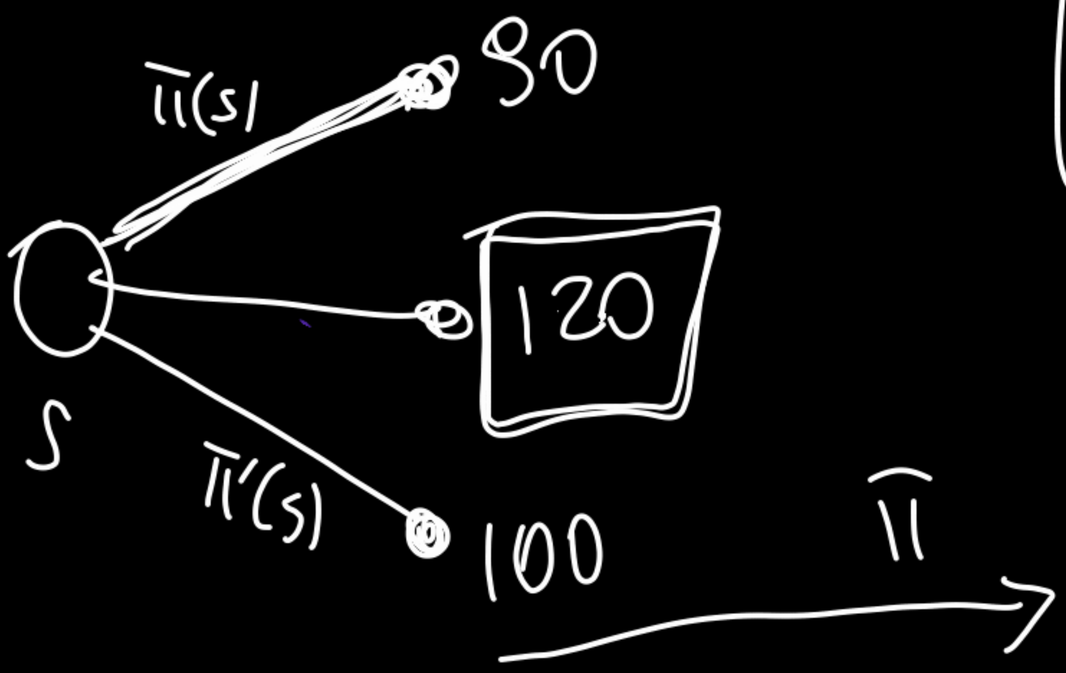
POLICY IMPROVEMENT (P.I.) — (DETERMINISTIC POLICIES).

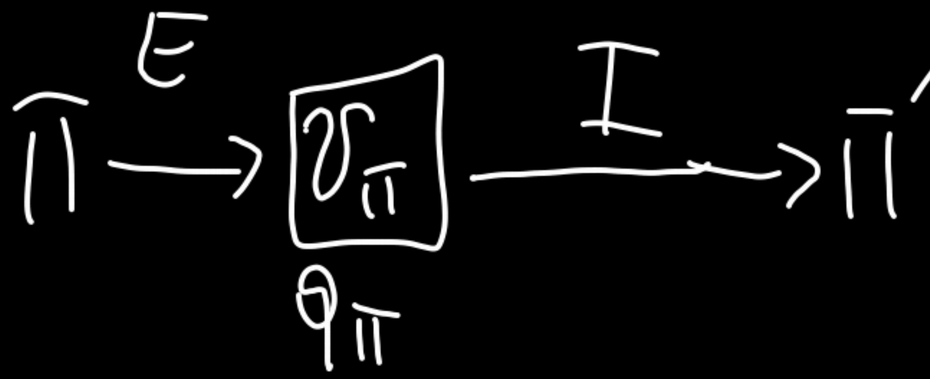
P.I. THEOREM: GIVEN POLICIES π', π, IF:

$$q_\pi(s, \pi'(s)) \geq v_\pi(s) \;, \forall s \rightarrow \pi' \geq \pi$$

POLICY IMPROVEMENT

$$\pi'(s) = \underset{a}{\operatorname{argmax}} \{ q_\pi(s, a) \}, \forall s$$





POLICY ITERATION (P.I)

$$\pi_0 \xrightarrow[(\infty \text{ STEPS})]{E} v_{\pi_0} \xrightarrow{I} \pi_1 \xrightarrow[(\infty \text{ STEPS})]{E} v_{\pi_1} \longrightarrow \cdots \longrightarrow v_{\pi^*} \rightarrow \pi^*$$

P.I WILL EVENTUALLY TERMINATE AND RETURN v_{π^*}, π^*.

IN FINITE MDP, FINITE NUMBER OF (DETERMINISTIC) POLICIES. (HOW MANY?)

CONSIDER POLICY IMPROVEMENT:

$$\pi \xrightarrow{I} \pi' \xrightarrow{E} v_{\pi'}$$

v_π

IN MOST CASES $\pi' > \pi$ π' IS A NEW POLICY

($v_{\pi'}(s) \geq v_\pi(s)$ $\forall s$, $v_{\pi'}(s) > v_\pi(s)$ FOR SOME s.)

AFTER SOME ITERATIONS

$v_{\pi'}(s) = v_{\pi}(s), \forall s$ STOP IMPROVING $\longleftrightarrow \pi' = \pi = \pi^*$ OPTIMAL.

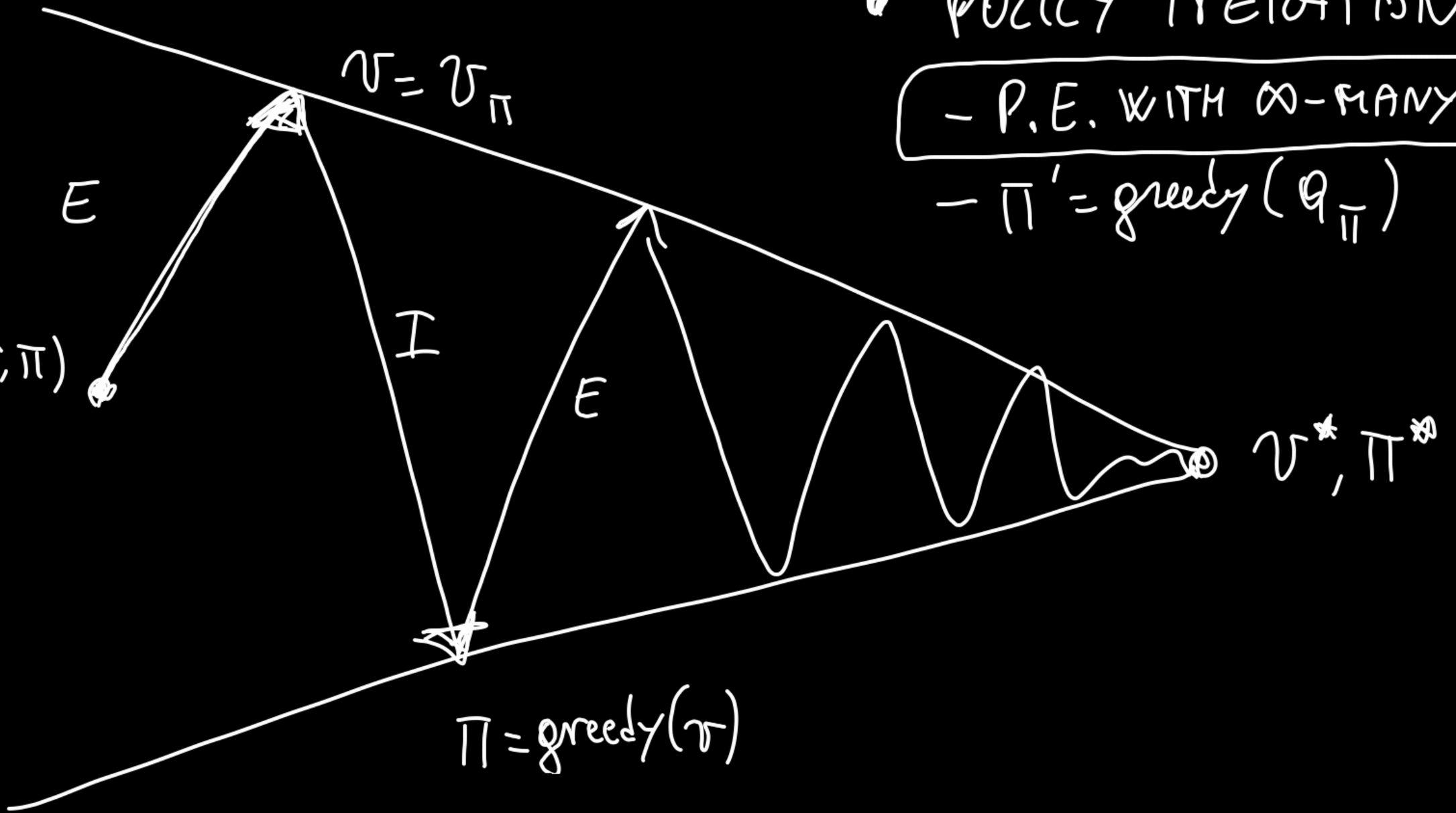
POLICY IMPROVEMENT

$$\pi'(s) = \arg\max_a \{ \underbrace{q_\pi(s,a)}_{\text{EXPAND}} \} = \arg\max_a \left\{ \sum_{s',r} p(s',r|s,a)\left[r + \gamma v_\pi(s')\right] \right\}$$

$$v_{\pi'}(s) = \max_a \underbrace{\left\{ \sum_{s',r} p(s',r|s,a)\left[r + \gamma v_{\pi'}(s')\right] \right\}}_{q_\pi(s,a)}$$

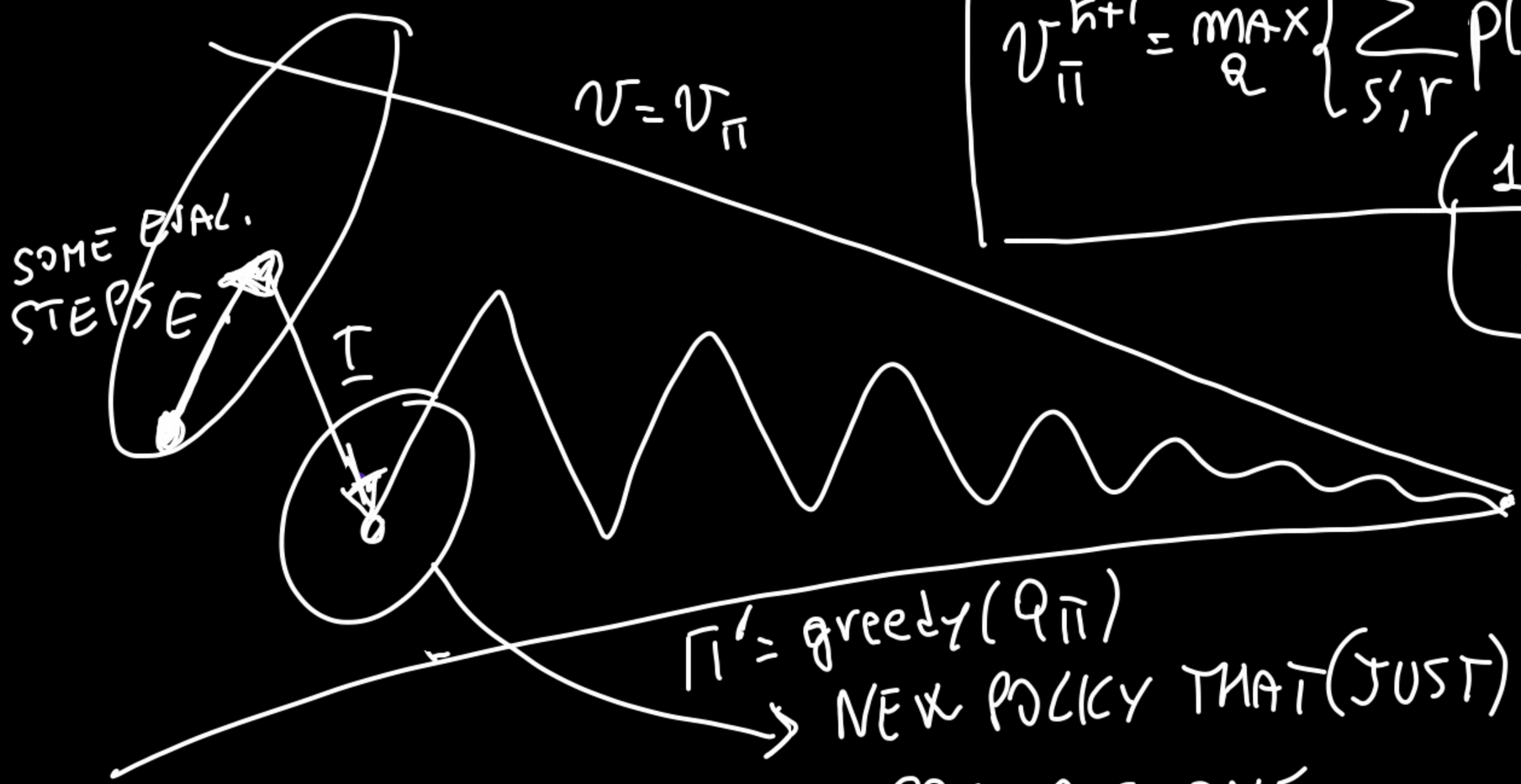
BELLMAN EQ. FOR v^*.

- POLICY ITERATION
  - P.E. WITH ∞-MANY STEPS
  - $\pi' = \text{greedy}(q_\pi)$



GENERALIZED POLICY ITERATION (G.P.I)

PRACTICAL ALGORITHMS THAT RELAX ONE OR BOTH STEPS OF P.I.



VALUE ITERATION

$$v_\pi^{k+1} = \max_a \left\{ \sum_{s',r} p(s',r|s,a)\left[r + \gamma v_\pi^k(s')\right] \right\}$$

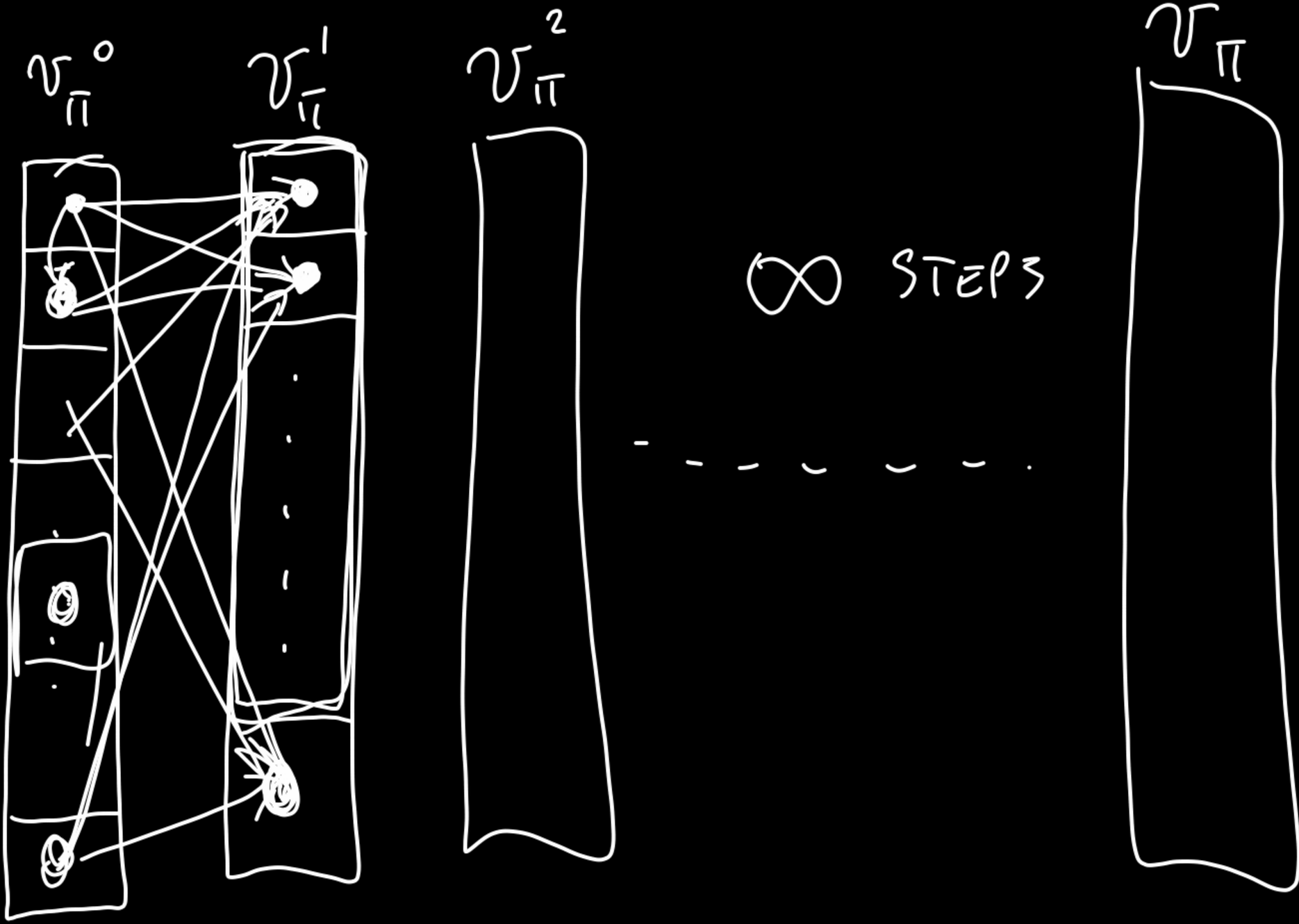
(1 EVALUATION STEP)

$$v_\pi^k \xrightarrow[k\to\infty]{} v^*(s)$$

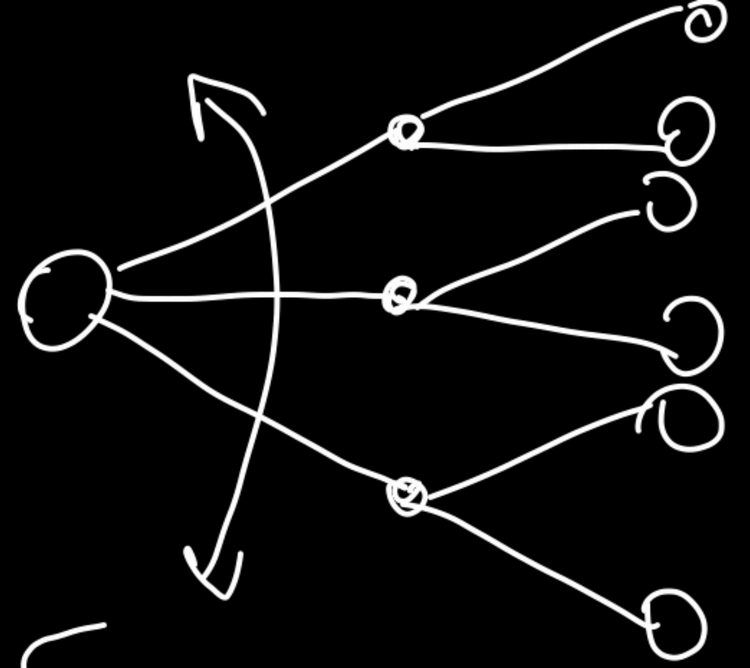
$\pi' = \text{greedy}(q_\pi)$

→ NEW POLICY THAT (JUST) IMPROVES THE PREVIOUS ONE

P.E. ($\pi \rightarrow v_\pi$)



DYNAMIC PROGRAMMING
- EXPECTED UPDATE



BOOTSTRAP METHOD

NEW ESTIMATE OF $v_\pi^{k+1}(s)$ IS AN AVERAGE OF PREVIOUS ESTIMATE OF VALUE OF SUCCESSOR STATES $v_\pi^k(s')$

LEARNING OPTIMAL POLICIES IN UNKNOWN ENVIRONMENT

($p(s', r | s, a)$ NOT KNOWN

- MONTE CARLO METHODS

- TEMPORAL DIFFERENCE

P, V ——>- EVACUATION (PREDICTION)
- IMPROVEMENT

PROBLEM.
WE CANNOT USE BELLMAN AS UPDATE RULE, SINCE p IS NOT KNOWN.

- EVALUATION (PREDICTION) IN MONTE CARLO (M.C.) METHODS.

$\pi \longrightarrow v_\pi$ WITHOUT KNOWING p

$$v_\pi(s) = E_\pi\{G_t \mid S_t = s\} \xrightarrow{DP} G_t = R_t + \gamma R_{t+1}$$

EPISODIC TASKS

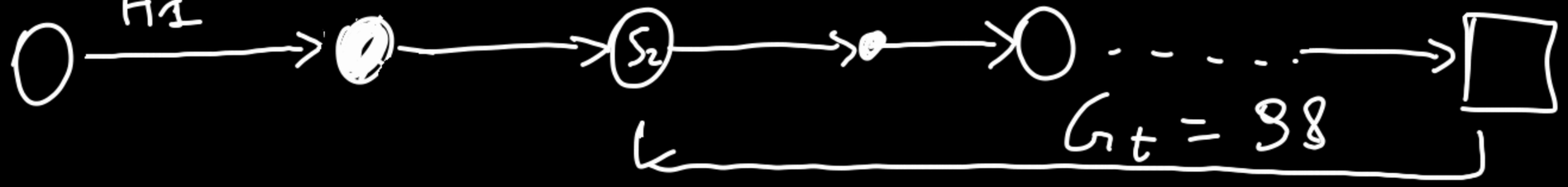
APPROXIMATE $v_\pi(s)$ AS THE AVERAGE (SAMPLE) OF THE RETURNS THAT WE GET IN MANY EPISODES, AFTER THE FIRST TIME WE VISIT S (FIRST-VISIT MC METHODS)

EPISODE (FULL INTERACTION WITH ENVIRONMENT, FROM START TO TERMINATION).

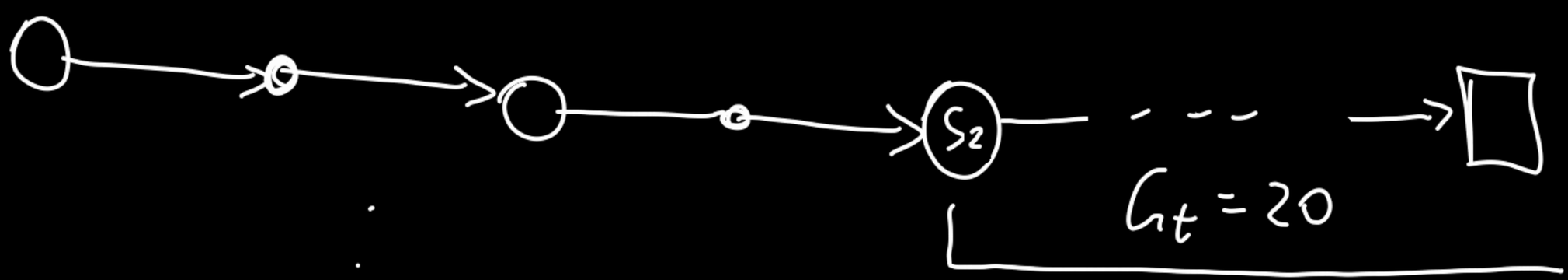
TRAJECTORY

HOW TO EVALUATE $v_\pi(s_2)$

EPISODE 1



// 2



WHEN NUMBER OF EPISODES $\longrightarrow \infty$ THE AVERAGES GO TO EXPECTED VALUE.

POLICY IMPROVEMENT

$$\pi'(s) = \arg\max_a \{ q_\pi(s,a) \}, \forall s$$

(IF P IS NOT KNOWN, THE GREEDY ACTION IS ALWAYS DEFINED WRT q)

$$= \arg\max_a \left\{ \sum_{s',r} p(s',r|s,a)\left[r + \gamma v_\pi(s') \right] \right\}$$

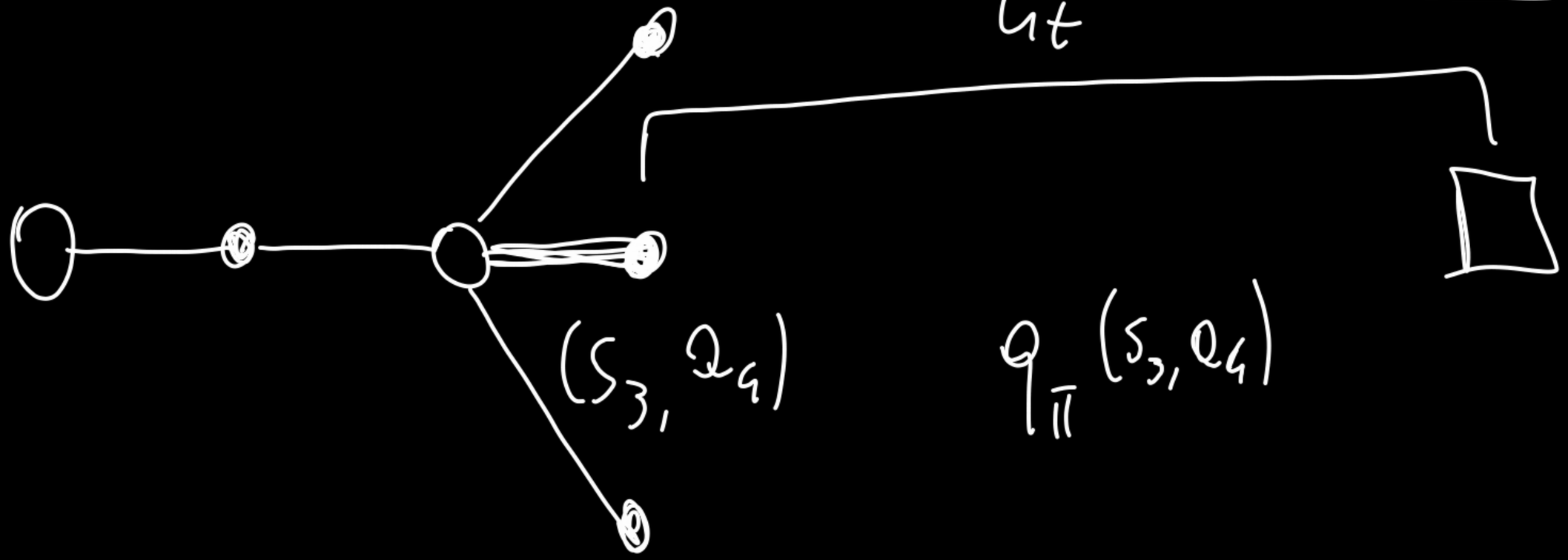
NOT KNOWN

SO WE NEED TO EVACUATE $q_\pi(s,a) = \mathbb{E}_\pi \{ G_t | S_t = s, A_t = a \}$

$q_\pi(s,a)$ IS ESTIMATED LIKE $v_\pi(s)$, BY DOING AVERAGES OF THE SAMPLE RETURNS OVER MANY EPISODES AFTER FIRST-VISIT (OR EVERY VISIT) OF STATE-ACTION NODES (s,a)

$q_\pi(s,a), \forall s,a$

G_t



TO ESTIMATE CORRECTLY $q_\pi(s,a)$, EVERY (s,a) NODE SHOULD BE VISITED INFINITE TIMES (WHICH DOES NOT HAPPEN IN GENERAL — E.G. CONSIDER DETERM. POLICY).

POSSIBLE SOLUTIONS:

(1) <u>EXPLORING STARTS</u>: START EVERY EPISODE FROM A RANDOM (s,a) NODE. ⟶ EVERY (s,a) IS VISITED INFINITE-MANY TIMES AVERAGES ⟶ $q_\pi(s,a)$.

(REAL LIFE EXP. VS SIMULATIONS (WORLD MODELS)

LEARNING IN UNKNOWN ENV. P)

ALTERNATIVES TO EXPLORING STARTS.

(2) ONE POSSIBILITY IS TO CHANGE POLICY IMPROVEMENT TO ASSURE EXPLORATION

STANDARD P.I

$\pi'(s) = \underset{a}{\operatorname{argmax}} \{ q_\pi(s,a) \}$, ~~DETERMINISTIC~~ ~~$\forall s$~~ $\rightarrow$

ε - GREEDY (ε-SOFT)

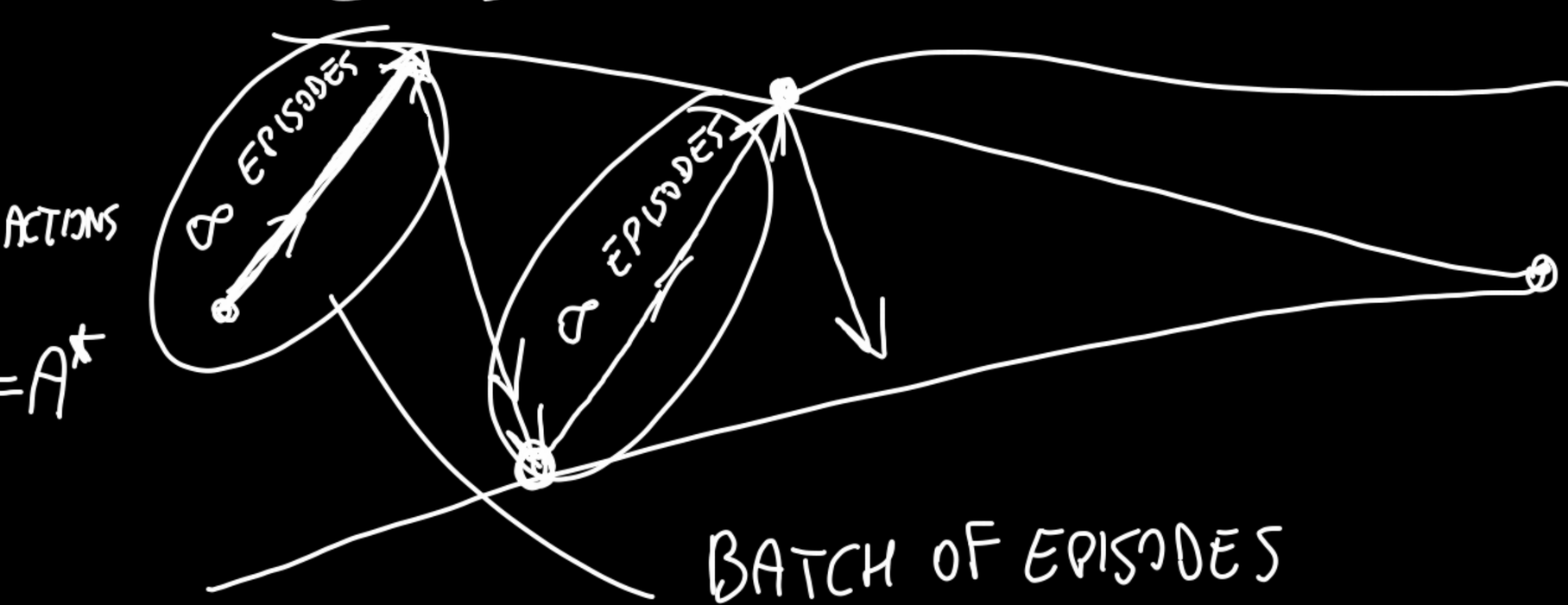
$A^*(s) = \underset{a}{\operatorname{argmax}} \{ q_\pi(s,a) \}$

$$\Pi(s,a) = \begin{cases} \dfrac{\varepsilon}{|A(s)|}, & \text{IF } a \neq A^* \\ 1 - \dfrac{\varepsilon}{|A(s)|}\overbrace{(|A(s)|-1)}^{\#\text{ OF NON-GREEDY}}, & \text{IF } a = \end{cases}$$

(ε-SOFT POLICY: $\pi(a|s) \geq \frac{\varepsilon}{|A(s)|}$)

- GRADUALLY WE CAN REDUCE ε, TO GET CLOSER TO THE DETERMINISTIC OPTIMAL POLICY.

CONVERGE TO THE BEST-POSSIBLE ε-SOFT POLICY.



GPI FOR MONTE CARLO.

- AFTER EVERY EPISODE, WE CAN UPTATE $V_\pi(s)$ FOR EVERY STATE VISITED IN THE EPISODE (SINCE WE HAVE THE SAMPLE RETURNS FROM THOSE STATES)

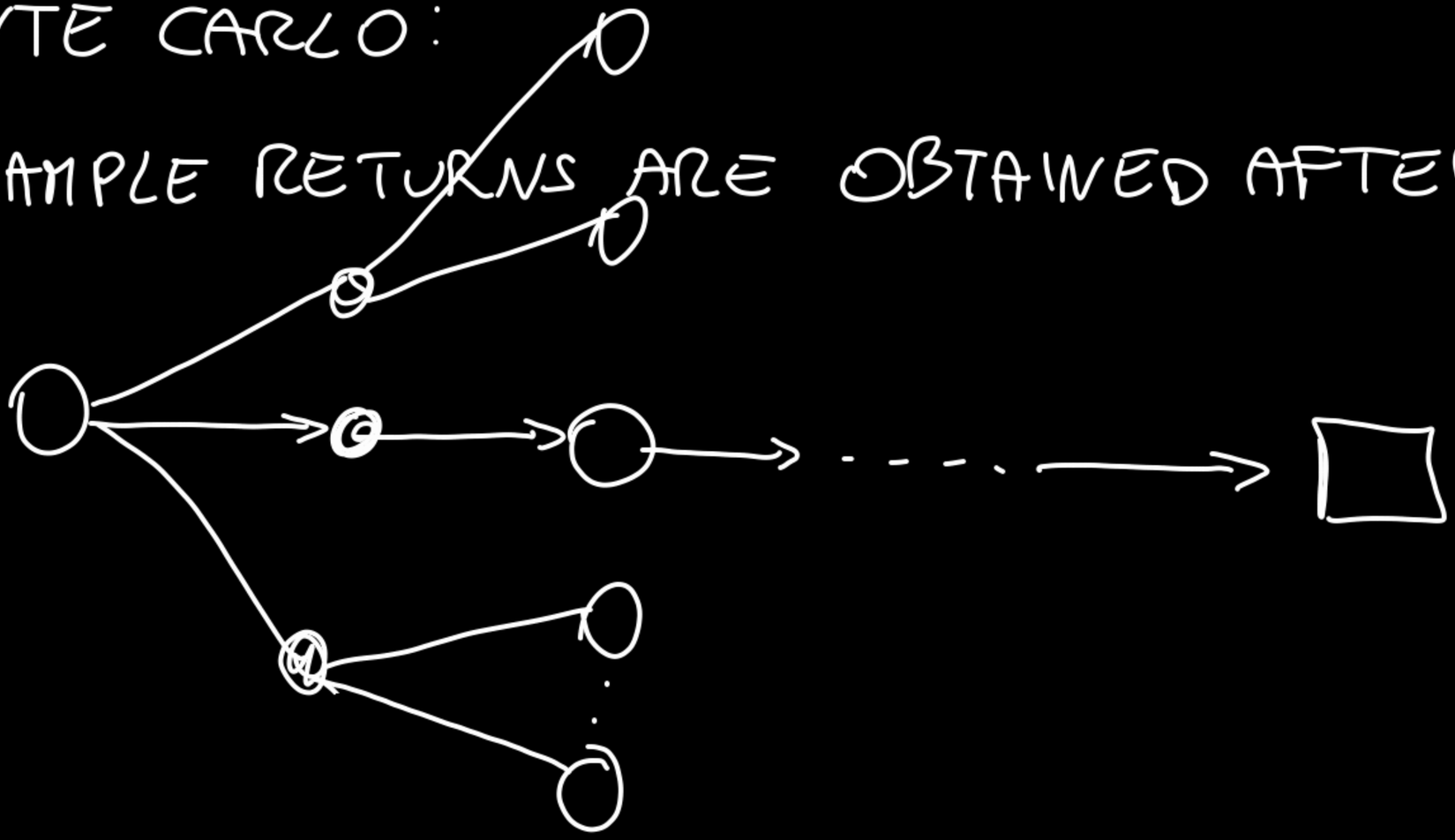
ITERATIVE AVERAGE

$$V(s) \leftarrow V(s) + \alpha[G - V(s)] \quad \forall s \text{ VISITED IN THE EPISODE}$$

G: SAMPLE RETURN

MONTE CARLO:

- SAMPLE RETURNS ARE OBTAINED AFTER FULL EPISODES



AT TERMINATION WE GET SAMPLE RETURNS THAT WE CAN USE TO UPDATE.

- NOT A BOOTSTRAPPING METHOD: AVERAGES OF RETURNS FROM FULL EPISODES.

THIRD POSSIBILITY TO ENSURE EXPLORATION:

- ON-POLICY RL ALGORITHMS: TRY TO IMPROVE THE POLICY THEY USE TO TAKE ACTIONS (TO EXPLORE ENV.).

- OFF-POLICY ALGORITHMS: LEARN THE OPTIMAL POLICY BY USING DATA COMING FROM ANOTHER POLICY.

  - ONE POLICY TO EXPLORE (BEHAVIOR POLICY π_b)

  - A SECOND POLICY (TARGET POLICY) THAT WILL CONVERGE TO THE OPTIMAL ONE.

POLICY EVALUATION (PREDICTION) FOLLOWING OFF-POLICY APPROACH.

π - TARGET POLICY (WE WAT THIS TO GO TO π^*)

π_b - BEHAVIOR POLICY (WE USE IT TO GENERATE DATA

TO ESTIMATE $v_\pi(s)$

$$\pi \xrightarrow{E.} v_\pi(s) \triangleq E_{\pi}\{G_t | S_t = s\}$$

HOW TO DO THIS USING DATA COMING FROM π_b ?

EXPERIMENTS ARE DONE FOLLOWING π_b.

- AT THE END OF EACH EPISODE, WE SHOULD "ADJUST" THE OBTAINED SAMPLE RETURNS,

